

CardioAI- A Machine Learning-Based Early Prediction System for Heart Disease

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Abstract—Heart disease continues to be one of the most frequent causes of death internationally, as early intervention is a strong factor in decreasing rates of mortality. This study includes CardioAI, an early prediction system for heart disease utilizing machine learning on clinical data of age, blood pressure, cholesterol, and ECG. The Cleveland Heart Disease dataset compose of 303 patients, utilized 14 attributes for analysis. Three models, including a Logistic Regression, Random Forest, and XGBoost were trained, and evaluated based on accuracy, precision, recall, F1, and ROC-AUC. The results concluded that Random Forest and XGBoost models considerably outperform Logistic Regression Models because they have higher accuracy and generalization potential. This study showcased how machine learning can aid healthcare professionals in a resource efficient and reliable way predicting cardiac risk factors.

Keywords—

1. Heart Disease
2. Machine Learning
3. Logistic Regression
4. Random Forest
5. XGBoost
6. Predictive Analytics.

I. INTRODUCTION

Heart disorder is a leading cause of death worldwide, making it a high priority for medical investigation and innovations. Diagnosis has typically relied on a clinical assessment, often requiring manual interpretation, usually with a delay and with greater subjectivity and human error. With the increase in health data and the advent of artificial intelligence (AI), machine learning (ML) models have the potential to revolutionize the diagnosis of heart disease through automated and early detection.

CardioAI, the combined predictive model proposed in this paper, employs supervised learning algorithms for predictive modeling, which can predict the risk of heart disease based on patient data. The aim is to develop an interpretable yet robust model to provide clinical professionals with the ability to efficiently screen and identify patients who are likely at risk.

CardioAI uses three main models: Logistic Regression, for baseline comparison; Random Forest, for ensemble-based classification; and lastly, XGBoost. For model evaluation, we will present the models with a number of different evaluation metrics, including accuracy, precision, recall, F1-score, and ROC-AUC.

II. LITERATURE

Many researchers have implemented machine learning for predicting heart disease on the Cleveland dataset and other cardiovascular datasets.

Lamir et al. [1] performed a comparative study on Logistic Regression, K-Nearest Neighbor (KNN), and Random Forest classifiers. Their Random Forest classifier produced 91% accuracy and an F1 score of 0.89, indicating that ensemble learning outperforms simple linear models.

Naik et al. [2] proposed a new framework based on Random Forest and XGboost that utilizes Social Group Optimization (SGO). They reported 95.08% accuracy with 95.26% ROC-AUC on the Cleveland dataset, providing evidence of evolutionary optimization's benefits when used for hyper-parameter tuning.

Teja et al. [3] implemented a variety of ML algorithms including Random Forest, AdaBoost, and XGBoost, to demonstrate the robustness of ensemble models. Their cross-validation metrics indicated achieving accuracy of 94% and 93% with Random Forest and XGBoost, respectively.

Kumar et al. [4] conducted an extensive review of machine learning applications for heart disease detection and suggested that feature engineering and interpretability are paramount for developing accurate and reliable models.

Alwakid et al. [5] developed an optimized XGBoost-based framework that incorporated feature selection techniques such as information gain and Chi-square and reported achieving up to 99% accuracy, while also being able to maintain interpretability.

Gaber et al. [6] proposed a federated learning architecture, FedCVD++, which Forest, and XGBoost to assist in privacy-preserving cardiovascular risk prediction, achieving a federated F1-score of 0.80.

Otoum and Nayak [7] suggested a framework employing differential privacy that prevents the observation of sensitive patient data while retaining a prediction accuracy of 85%, thereby managing the trade-off between model prediction performance and data confidentiality.

Gap Identified:

These contributions have demonstrated federated learning models can predict cardiovascular risk using sensitive patient data while retaining confidentiality. However, the proposed systems tend to focus on optimizing for accuracy by applying complex architectures resulting in reduced interpretability. This project will address the gaps by building an interpretable and trustworthy model using simple transparent algorithms, with strong regression predictive performance.

III. PROPOSED WORK

The conceptualized prototype, CardioAI, intends to assess the probability of heart illness in patients using machine learning models fitted on medical data. The aim is to create a highly interpretable, efficient, and accurate model that can assist healthcare professionals with earlier diagnosis and in preventative healthcare.

To further develop the framework, we have organized the system architecture schematic (Fig. 1) into four main stages: Data Acquisition, Data Preprocessing, Model Development, and Performance Evaluation. The first stage of the framework pipeline begins as a collection of patient medical records from the Cleveland Heart Disease dataset. Following the collection, the next stage is to clean and transform the data to prepare it for training our model. Next, the cleaned data will be divided into training and testing data splits which will later be used to train the model and evaluate the performance of the model for generalization, as well as assess an unbiased performance.

The framework puts for three machine-learning algorithms - Logistic Regression, Random Forest, and Extreme Gradient Boosting (XGBoost) - to predict the risk of heart disease. Logistic Regression will be a baseline model to allow for interpretability and is a simple model. Random Forest is a method based on ensemble learning, reducing variance and increasing the reliability of the prediction with multiple decision trees. XGBoost is a method based on gradient boosting and is focused on measuring classification performance by gradually minimizing classification error based on boosting.

Hyperparameter optimization techniques, including Randomized Search and Grid Search, are used to tune the hyperparameters of each of the models. The models are evaluated on standard classification metrics: Accuracy, Precision, Recall, F1-Score, and the ROC-AUC value. The model with the highest balance in precision and recall is considered the best predictor of risk for heart disease.

Ultimately, the goal of CardioAI is to create a transparent and actionable prediction framework that can be incorporated into both clinical decision-support systems. By assessing the framework's capacity to generalize on unseen data, the CardioAI model can be a useful tool in identifying patients for early vitamin lifestyle and medical intervention, thus aiding medical practitioners in early detection of high-risk patients as well as subsequent intervention planning.

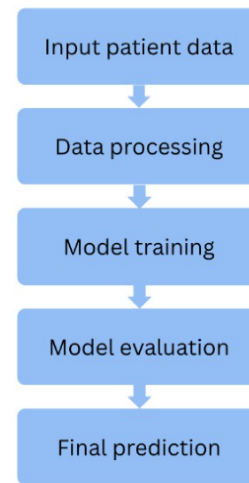


FIG 1. BLOCK DIAGRAM

IV. METHODOLOGY

A. Dataset Collection

The dataset chosen for this research is the Cleveland Heart Disease Dataset, received from the UCI Machine Learning Repository. This dataset has a total of 303 patient records and includes 14 clinical attributes like age, sex, type of chest pain, resting blood pressure, serum cholesterol, fasting blood sugar, resting electrocardiographic results (ECG), maximum heart rate achieved, and the target output variable. The target variable indicates the presence or absence of heart disease, i.e., a value of 1 refers to heart disease patients and 0 refers to healthy individuals. This dataset is selected due to its wide variety of applications in previous research work; hence, this was deemed relevant for this study of evaluating classification models for heart disease prediction.

B. Data Preprocessing

Before model learning, the dataset was loaded into a specific preprocessing pipeline that ensured appropriate quality and consistency on the data. For this, missing values represented as "?" or "-9" were replaced with the median values for their respective attributes to reduce data distortion. The continuous variables like cholesterol and blood pressure had their values scaled to a common range using the StandardScaler in Scikit-learn to maintain a leveled baseline and reduce bias in the model training. Categorical features, including chest pain type and thalassemia, were turned into numerical values using one-hot encoding so that they could be compatible with the algorithms that solve machine learning problems.

Next, the dataset was split into training and test subsets in an 80:20 ratio using stratified sampling. This practice ensured that the class distribution for the training and test subsets were consistent to avoid bias in the model toward the majority of the class distribution and enabled fair performance assessment.

C. Model Building

Three supervised learning models were created and compared in this study.

The first model is Logistic Regression, which served as the baseline model. Logistic Regression is a linear and interpretable approach, designed to estimate the likelihood of heart disease using the weighted input features.

The second model built using Random Forest, is an ensemble learning technique that creates multiple decision trees from random subsets of the data and features, and then combines the results by majority vote. This helps to lessen overfitting, while improving the model's generalization.

The third model built is called, Extreme Gradient Boosting (XGBoost). This is an optimized version of gradient boosting that builds sequential decision trees that each correct the mistakes of their predecessor. XGBoost is designed to handle missing data effectively, while also supporting regularization for overfitting, and tends to generate higher accuracy scores on tabular data sets.

D. Hyperparameter Tuning

To improve performance of the model, hyperparameter tuning was performed using Randomized Search and Grid Search techniques with cross-validation. The tuning of hyperparameters modified important parameters such as the number of estimators, maximum depth of trees, learning rate, and class weights. The best hyperparameter setting for the Random Forest model was 200 estimators, maximum depth of trees at 15, and balanced class weights. The tuning reduced any bias and variance trade-offs, with best performance similar across the folds of validation.

E. Evaluation Metrics

In order to thoroughly evaluate each model, standard classification metrics were used.

Accuracy measured the overall correctness of predictions.

Precision measured the number of true positives over the total predicted positives.

Recall (or Sensitivity) measured the number of true positives versus actual positives.

F1-Score is the harmonic mean of precision and recall - balancing the two.

ROC-AUC (Receiver Operating Characteristic - Area Under Curve) evaluated the model's ability to separate between the classes at different thresholds.

These metrics provide an aggregate view of the performance of each model and allow fair comparison and identification of the best approach to predict heart disease.

V. RESULT AND DISCUSSION

A. Model Performance

Three supervised learning models—Logistic Regression, Random Forest, and XGBoost—were trained and evaluated on the Cleveland Heart Disease dataset. Each model's performance was assessed on accuracy, precision, recall, F1-score, and ROC-AUC metrics.

Logistic Regression achieved an overall accuracy of 80.3% on the dataset. This model had a precision of 0.864 and recall of 0.679 on non-disease cases, and a recall of 0.909 and F1-score of 0.833 for detecting heart disease. It achieved a ROC-AUC score of 0.869, which indicates a good ability to distinguish between positive and negative for other diseases. This model performed as a strong baseline, showing balance in generalization between training and testing data without significant overfitting.

Slightly outperforming Logistic Regression, Random Forest had a test accuracy of 82.0%. Class 0 had a precision of 0.947, class 1 had a precision of 0.762, and recall for heart disease was 0.970. The F1-score reached 0.853, and ROC-AUC was 0.912 indicating a higher ability to classify and improved discrimination above logistic regression. It was also marked by 100% accuracy on the training dataset, which showed mild overfitting typical for ensemble tree-based methods.

With hyperparameter tuning, XGBoost showed comparable performance, achieving a test accuracy of 78.7% and AUC score of 0.915. The reconstruction achieved a recall of 0.909 for the heart disease class and precision of 0.857 for the non-disease class. Test accuracy was slightly less than the Random Forest, but the relatively high AUC and favorable precision-recall balance signaled the potential for good generalization. The model also achieved a training accuracy of 100%, indicating strong learning capability but would benefit significantly from regularization methods to further diminish overfitting.

B. Confusion Matrix Comparison

The confusion matrices served to reinforce the assessment of the models.

Logistic regression identified 30 positive cases and 19 negative cases. It misclassified 12 samples.

Random Forest included 32 true positives and 18 true negatives. It misclassified 11 samples.

The confusion matrix for XGBoost was similar with 32 true positives and 17 true negatives, along with misclassification of 12 samples.

Overall, the two ensemble based methods, Random Forest and XGBoost, performed better than logistic regression in terms of false negatives. This is significant for medicine forecasting systems since it may have consequences for missing positive cases.

C. ROC Curve Comparison

All models showed strong classification performance, with AUC scores of 0.869 for Logistic Regression, 0.912 for Random Forest, and 0.915 for XGBoost. XGBoost had the highest AUC, meaning it provides the best trade-off between sensitivity and specificity.

D. Comparative Discussion

Overall, Random Forest provided the best trade-off of accuracy and interpretability for heart disease prediction, with improved recall and F1-score compared to Logistic Regression, while still providing simplicity and transparency. While XGBoost marginally improved the ROC-AUC, this came at the cost of reduced accuracy due to a sensitivity to tuning parameters.

These findings are consistent with previous literature [3][4][5], which reported that ensemble learning techniques (e.g., Random Forest or XGBoost) consistently outperformed linear models in structured medical datasets. The findings confirm that CardioAI can provide early reliable heart disease prediction to aid clinical decision-making with interpretable machine learning models, while still providing high accuracy.

VI. CONCLUSION

In this research, CardioAI was introduced as a machine learning-based framework for early prediction of heart disease with the Cleveland Heart Disease dataset. Three models, Logistic Regression, Random Forest, and XGBoost, were developed and compared against their performance metrics. The findings indicated that ensemble learning models outperform linear classifiers in all dimensions of accuracy, recall, and AUC.

Of the three models, Random Forest achieved the best overall balance at 82% accuracy, 0.912 ROC-AUC, and strong recall of positive cases. XGBoost, as an alternative, achieved a small increase in discriminative power with a 0.915 AUC. While Logistic Regression is less powerful and only a baseline, it is simple enough for the reader to interpret and report the results.

The proposed system shows the benefits of using clear and data-based decision support tools in health care. By facilitating early identification of cardiac risk, these systems can aid clinicians in preventive diagnosis and improve patient outcomes. Future efforts should include a larger dataset from various sources, incorporate deep learning models, and explore explainable AI (XAI) techniques to enhance understanding and build trust in real-world clinical applications.

VII. REFERENCES

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